Day 054 | Unit 04 | Teacher plan

Numeric and Algebraic Expressions

Evaluate Expressions and Formulas

Standards: 6.EE.A.2c

Learning target: I can substitute values and evaluate expressions or formulas.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 5

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Replace every variable with its assigned value, using parentheses. Apply the order of operations. In formulas, label the answer with the appropriate measurement units.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 14

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01 / Learn

Replace every variable with its assigned value, using parentheses. Apply the order of operations. In formulas, label the answer with the appropriate measurement units.

Worked example 1

Evaluate 4x + 4 for x = 6.

1. Replace every variable with its assigned value, using parentheses.

2. Apply the order of operations.

3. Substitute first: 4 × 6 + 4 = 28.

4. State the result in the form requested: 28.

Answer: 28

Worked example 2

Let x = 5/2 and y = 5. Evaluate 4x + y^2. Part A: enter the value. Part B: show substitution and the order of operations.

1. Replace every variable with its assigned value, using parentheses.

2. Substitute: 4 × (5/2) + 5^2.

3. The product is 10 and the square is 25.

4. Add to get 35.

Answer: 35

Day 054 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Evaluate 8x + 5 for x = 2.

02 / Constructed response

Evaluate 8x + 6 for x = 2.

03 / Constructed response

Evaluate 4x + 6 for x = 3.

04 / Constructed response

Evaluate x^2 + 5y for x = 6 and y = 4.

05 / Constructed response

Evaluate x^2 + 3y for x = 3 and y = 3.

06 / Constructed response

Evaluate x^2 + 2y for x = 4 and y = 3.

07 / Constructed response

A rectangle has length x + 6 and width 4. Use A = lw to write its area expression and find A when x = 6.

08 / Numeric entry and reasoning

Let x = 3/2 and y = 5. Evaluate $3x + y^2$. Part A: enter the value. Part B: show substitution and the order of operations.

09 / Numeric entry and reasoning

Let x = 1 and y = 5. Evaluate $7x + y^2$. Part A: enter the value. Part B: show substitution and the order of operations.

10 / Constructed response

A rectangle has length x + 6 and width 6. Use A = lw to write its area expression and find A when x = 2.

Day 054 | Unit 04 | Answer key

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1. 21

Substitute first: 8 × 2 + 5 = 21.

6.EE.A.2c | Constructed response

2. 22

Substitute first: 8 × 2 + 6 = 22.

6.EE.A.2c | Constructed response

3. 18

Substitute first: 4 × 3 + 6 = 18.

6.EE.A.2c | Constructed response

4. 56

Substitute first: 6^2 + 5 × 4 = 56.

6.EE.A.2c | Constructed response

5. 18

Substitute first: 3^2 + 3 × 3 = 18.

6.EE.A.2c | Constructed response

6. 22

Substitute first: 4^2 + 2 × 3 = 22.

6.EE.A.2c | Constructed response

7. A = 4(x + 6); 48 square units

Substitute x = 6: 4(6 + 6) = 48.

6.EE.A.2c | Constructed response

8. 59/2

Substitute: 3 × (3/2) + 5². The product is 9/2 and the square is 25. Add to get 59/2.

6.EE.A.2c | Numeric entry and reasoning

9. 32

Substitute: 7 × (1) + 5². The product is 7 and the square is 25. Add to get 32.

6.EE.A.2c | Numeric entry and reasoning

10. A = 6(x + 6); 48 square units

Substitute x = 2: 6(2 + 6) = 48.

6.EE.A.2c | Constructed response